Unit: 6 Unit 5: Linear Equations **Name:** ________________

Topic: Scatter Plot and The Best Fit Line **Date:** ___________

Foundation (Jalapeno)

Q1. What is the purpose of a scatter plot?
(A) To show correlation
(B) To show causation
(C) To show regression
(D) To show classification
(E) To show clustering

Q2. Which type of correlation is shown by the data points $(1, 2)$, $(2, 4)$, $(3, 6)$?
(A) Positive
(B) Negative
(C) None
(D) Weak
(E) Strong

Q3. What does the line of best fit represent?
(A) The average of the data points
(B) The median of the data points
(C) The trend of the data points
(D) The correlation of the data points
(E) The regression of the data points

Q4. If the data points $(1, 2)$, $(2, 3)$, $(3, 5)$ have a line of best fit $y = 2x - 1$, what is the predicted value of y when $x = 4$?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q5. Which of the following is a characteristic of a positive correlation?
(A) As x increases, y decreases
(B) As x decreases, y increases
(C) As x increases, y increases
(D) As x decreases, y decreases
(E) No relationship between x and y

Q6. What is the equation of the line of best fit for the data points $(1, 1)$, $(2, 3)$, $(3, 5)$?
(A) $y = x + 1$
(B) $y = 2x - 1$
(C) $y = x - 1$
(D) $y = 2x + 1$
(E) $y = x^2$

Q7. If the data points $(1, 2)$, $(2, 4)$, $(3, 6)$ have a correlation of 1, what can be said about the relationship between x and y?
(A) There is a weak positive correlation
(B) There is a strong positive correlation
(C) There is a weak negative correlation
(D) There is a strong negative correlation
(E) There is no correlation

Q8. What is the purpose of drawing a line of best fit?
(A) To show the average of the data points
(B) To show the trend of the data points
(C) To show the correlation of the data points
(D) To show the regression of the data points
(E) To show the classification of the data points

Open Ended Questions (FRQ)

Q9. Given the data points $(1, 2)$, $(2, 3)$, $(3, 5)$, create a scatter plot and draw the line of best fit. Then, use the line of best fit to predict the value of y when $x = 4$. Show your work and explain your reasoning.

Q10. Explain the difference between a positive correlation, a negative correlation, and no correlation. Provide examples of each and explain how the line of best fit is used in each case.

Chef's Tips

- Tip 1: Ensure students understand the concept of correlation and how it relates to the line of best fit.
- Tip 2: Emphasize the importance of showing work and explaining reasoning in free-response questions.
- Tip 3: Remind students to use proper mathematical notation and formatting in their responses.